

Exercise Sheet 3

Aufgabe 3.1: Interest Rates

$$FV = PV(1 + r)^t$$

$$PV = \frac{FV}{(1 + r)^t}$$

FV = value in the future

PV = value today / initial investment

r = annual interest rate as a decimal

t = number of years

$$PV = C \cdot \frac{1 - (1 + r)^{-t}}{r} + \frac{F}{(1 + r)^t}$$

PV = present value of the bond

C = yearly dividend/coupon payment

r = market interest rate

t = number of years

F = face value / original investment paid back at the end

- (a) Sofia's investment of 100€ grows at a 10% interest rate p.a. How much money does she have after 8 years?

$$FV = 100(1 + 0.1)^8 = 214.36\text{€}$$

(b) You buy a bond for 100€ and in return get 10€ in dividends for ten years. After ten years, you also get the original investment back. At a market rate of 8%, what is the Present Value of the bond?

$$PV = 10 * \frac{1 - (1 + 0.08)^{-10}}{0.08} + \frac{100}{(1 + 0.08)^{10}} = 113.42\text{€}$$

Aufgabe 3.2: Probability of Default

```
library(ggplot2)

df <- data.frame(company=c("Star", "Cow", "Dog"),
                 interest_rate=c(5.20, 5.40, 9.00))
head(df)

# Basic barplot
p<-ggplot(data=df, aes(x=company, y=interest_rate)) +
  geom_bar(stat="identity", fill="steelblue")+
  geom_text(aes(label = paste0(interest_rate, "%")), vjust=1.6, size=3.5,
color="white")+
  labs(
    x = "Company",
    y = "Interest Rate (%)"
  )
p

minimal_risk_for_surcharge = function(r_free, prob_default, recovery_rate) {
  (1 + r_free) * (prob_default*(1-recovery_rate))/(1-prob_default*(1-recovery_rate))
}

data1 = data.frame(company=c("Star", "Cow", "Dog"),
                  prob_default=c(0.03,0.12,7.27),
                  r_premim=c(0.02,0.09,5.00),
                  agreedinterest=c(5.20,5.40,9.00),
                  r_min=c(0,0,0),
                  npv=c(0,0,0)
                  )

r_free = 0.049
recovery_rate = 0.3945
market_rate = 5/100

data1$r_min = mapply(minimal_risk_for_surcharge, r_free,
data1$prob_default/100,recovery_rate)*100
```

```

npv_computal = function(today_cashflow, agreedinterest, market_rate,
risk_premium) {
  today_cashflow + (abs(today_cashflow) *
(1+agreedinterest))/(1+market_rate+risk_premium)
}

data1$npv = mapply(
  npv_computal,
  today_cashflow = -1000000,
  agreedinterest = data1$agreedinterest / 100,
  market_rate = market_rate,
  risk_premium = data1$r_premim / 100
)

print(paste0("The company with highest NPV: ",
data1$company[which.max(data1$npv)]))

```

Aufgabe 3.3: EU Population

```

library(tidyverse) # Load Library

df = read.csv("eu_population.csv", check.names = FALSE) #load csv into df variable,
without changing anything

df_tidy = df %>%
  pivot_longer(
    cols = !Country,
    names_to = "Year",
    values_to = "Population"
  )

# Take df and send it(pipe it) into the function pivot longer
# Takes all columns except Country and turn them into rows
# old columns will go into a new column named Year
# the numbers inside those columns will go to a new column Population

df_tidy %>%
  filter ( Year == max(Year) ) %>%
  arrange ( desc(Population)) %>%
  slice_head ( n = 6 )

# Show first 6 rows, sorted in descending order of the maximum year

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```
df_tidy %>%  
  filter ( Country == "France" | Country == "Poland") %>%  
  filter ( between(as.numeric(Year), 2018, 2020) ) %>%  
  arrange ( desc(Year))  
# Shows information about France and Poland in range 2018-2020, sorted by year in  
descending order
```